Daedalus Calibrations and Usage

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Calibrations & Usage of the Daedalus Projected Field Magnet

The methodology used in getting the requested field for the projected field magnet is arcane so we explain it here.

The most recent center calibration was ran in September 2018. The most recent strength calibration was ran a while ago, 2017 or before.

Determining Field Orientation

- The field center cross on the setup is always kept along the central axis of the magnet, defined as the line normal to the plane of the magnet with the GMW sticker on it. We call this axis the y axis since when phi=0 this is along the y axis of the Newport motion stage.
- There is a particular calibration for the X and Y coordinates of the motion stage which sets the "center" of the magnet at the field center on the setup.
- When the magnet is in this position, the field produced at the cross is entirely in-plane and the azimuthal angle phi can be set by rotating the magnet.
- To set an out-of-plane polar angle, the magnet is moved along the y axis which causes the fringing part of the field to create an out-of-plane component of the magnetic field. There is a calibration which gives the radial distance you must move the magnet out to get the desired polar angle.
- Moving in the positive y direction creates a field with an upwards out-of-plane component, and by symmetry moving in the negative y direction creates a field flipped across the xy-plane.
- In the calibration, the polar angle is defined such that a completely in-plane field has a polar angle of zero and an upwards directed field completely out-of-plane has a polar angle of 90 deg.
- Another quirk is that the fitting polynomial giving the radial distance *must* take a negative number for the polar angle.

Negative Fields and Angles

- For a positive field, assuming phi=0, moving in +y yields a B vector pointing upwards (away from the magnet) and moving in -y produces a B vector pointing downwards (towards the magnet) as expected by symmetry.
- If a negative field is requested, the magnet moves to the same physical position as it would if a positive field was being requested but runs current in the opposite direction. Therefore, with a negative field strength the magnetic field's polar angle will effectively change sign and its azimuthal angle will rotate by 180 deg.
- If a negative polar angle with a positive field is requested, the magnet moves the same radial distance it would for the corresponding positive polar angle but in the opposite direction. This keeps the azimuthal angle in the expected direction.

Determining Field Strength

There are two steps to determining the amount of current to supply to the magnet to get a desired field strength.

1. The base voltage calibration directly returns the voltage output from the DAQ (which controlls the magnet's power supply) as a function of desired field strength, assuming the magnet is centered on the mark on the setup (i.e. magnetic field completely in-plane).

2. If an out-of-plane field is requested there is another calibration, a function of the magnitude of the radial distance moved, which gives a correction to the voltage. The voltage from the base calibration is divided by this correction to give the appropriate voltage to give the desired field strength.

Strength Limitations

Because our control voltage to the power supply cannot exceed 10V, there is a limit to the total magnetic field strength which depends on the radial distance moved. Because the radial distance moved also affects the polar angle of the field, there is a somewhat complicated dependence on the maximum in-plane and out-of-plane field we can apply as a function of the polar angle requested. A plot summarizing the limit with the most recent calibrations is shown below.

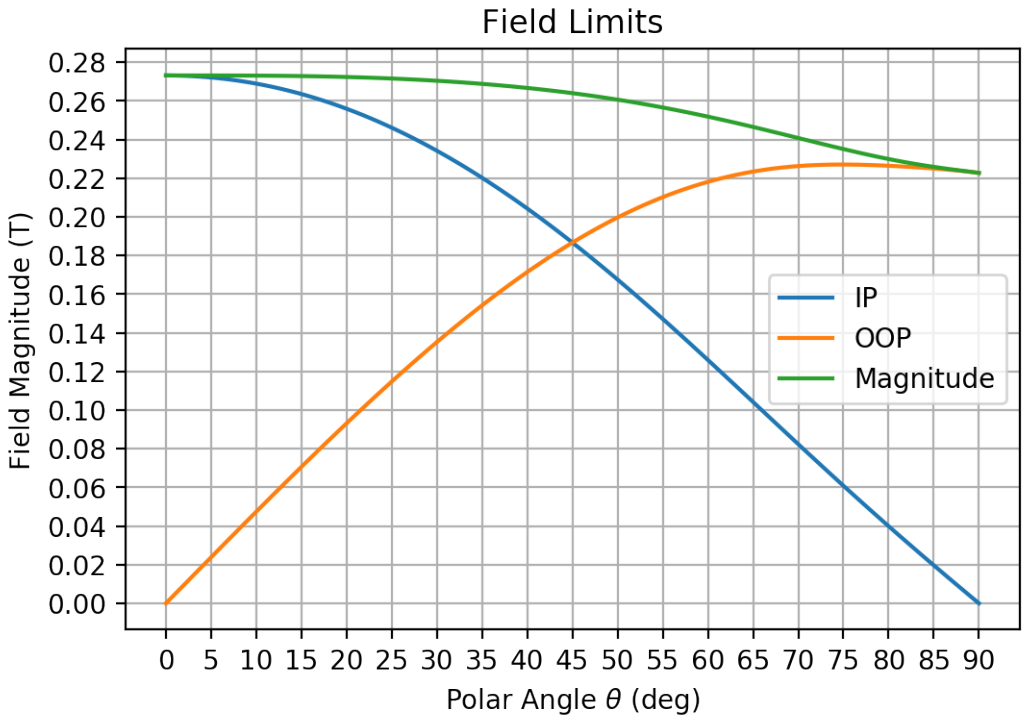


Figure 1: Maximum magnetic field calibration

A more detailed look at the field strength calibrations, as well as how to use these calibrations, is shown in the Jupyter notebook `calibration_example.ipynb`.

Usage Notes

Some tips for programming the magnet effectively: - Moving the magnet is very time consuming in experiments. So, it is usually better to have anything dealing with moving the magnet as the "outermost loop" of your measurements. So if you are doing an STFMR measurement and want to sweep dc bias, frequency, and azimuthal angle the angle sweep should be the "outer loop." - If you want to sweep the out of plane field and have an in-plane component be static, the best way to do this is to keep the field sign positive always and return negative polar angles. The in-plane component of the field will be change the least in this way.

Working with (or Against) the ESP300 Motion Controller

The Newport ESP300 motion controller has a difficult time dealing with GPIB commands. We have enumerated some tips to keep in mind if you ever need to write control code for it: - Motion not done when it says it is, add

extra delay - Frequently gives PCIe timeout, should catch and re-try instead of having whole application crash - Should check that you've gotten to the expected position, sometimes does not actually go where it says it has. Noticed mostly with phi axis. - Check for errors often to clear the error buffer. - Prepare for non-int error return codes - Should try not to send commands too quickly, it can get confused

Limitations of the Current Calibration

As it stands, the calibration has some limitations when it comes to applying an out-of-plane field. To achieve a completely out-of-plane field, the magnet must move a radial distance of 14.4 mm from the center of the calibration. However, the center of the calibration is at (12.25,13.54) so clearly there will be some azimuthal angles at which this will not work. Although for a completely out-of-plane field this would not be an issue, for angles with a large polar angle this could be a problem. With the current calibration, the largest polar angle we can apply and have a full range of movement in the azimuthal direction is 81 deg. We have supplied a sketch of the setup showing the problem. The circle in the diagram represents the circle we would need to move around to produce a completely out-of-plane field. All measurements are in mm.

Fixing the Motion Controller when Things Go Wrong

The Newport ESP300 is a hot pile of garbage and can get stuck easily. Here is a list of ways we have seen things go quite wrong and how we can fix them

Center Calibration Seems Wrong

Whenever the ESP300 is power-cycled, wherever the XY stages are when the instrument is powered back on is where it will set the software zero. Since we assume that zero is at the negative hardware limits of the motion stages, the magnet will move to the wrong position. Instructions for fixing this are given below. Values are given consistent with the most recent calibration.

Before starting any measurements, you should check that the magnet is: in about the right position. There are two things to check:

- Check that the reading on the screen of the Newport is X=12.25, Y=13.54
- Check that the platform is actually in the right position. The X and Y stage of the magnet's motion controller are each made of two plates, one base plate on the bottom and another movable plate on top. When the magnet is appropriately zeroed, these two plates should be right on top of one another. If something is wrong they will probably be off by ~1cm.

If either of these things are wrong, you must reset the magnet. instructions for that are below.

Re-zeroing the magnet

1. Open the Anaconda prompt
2. type `python` to open a Python interpereter
3. Type the following commands in the interpreter:

   ```
   from pymeasure.instruments.newport import ESP300
   a = ESP300(10)
   a.x.position = -100
   a.y.position = -100
   ```

   after this, wait for the Newport to stop moving. It should beep to let you know that the X and Y stages have reached their negative hardware limits.
4. Now set the proper software zero:

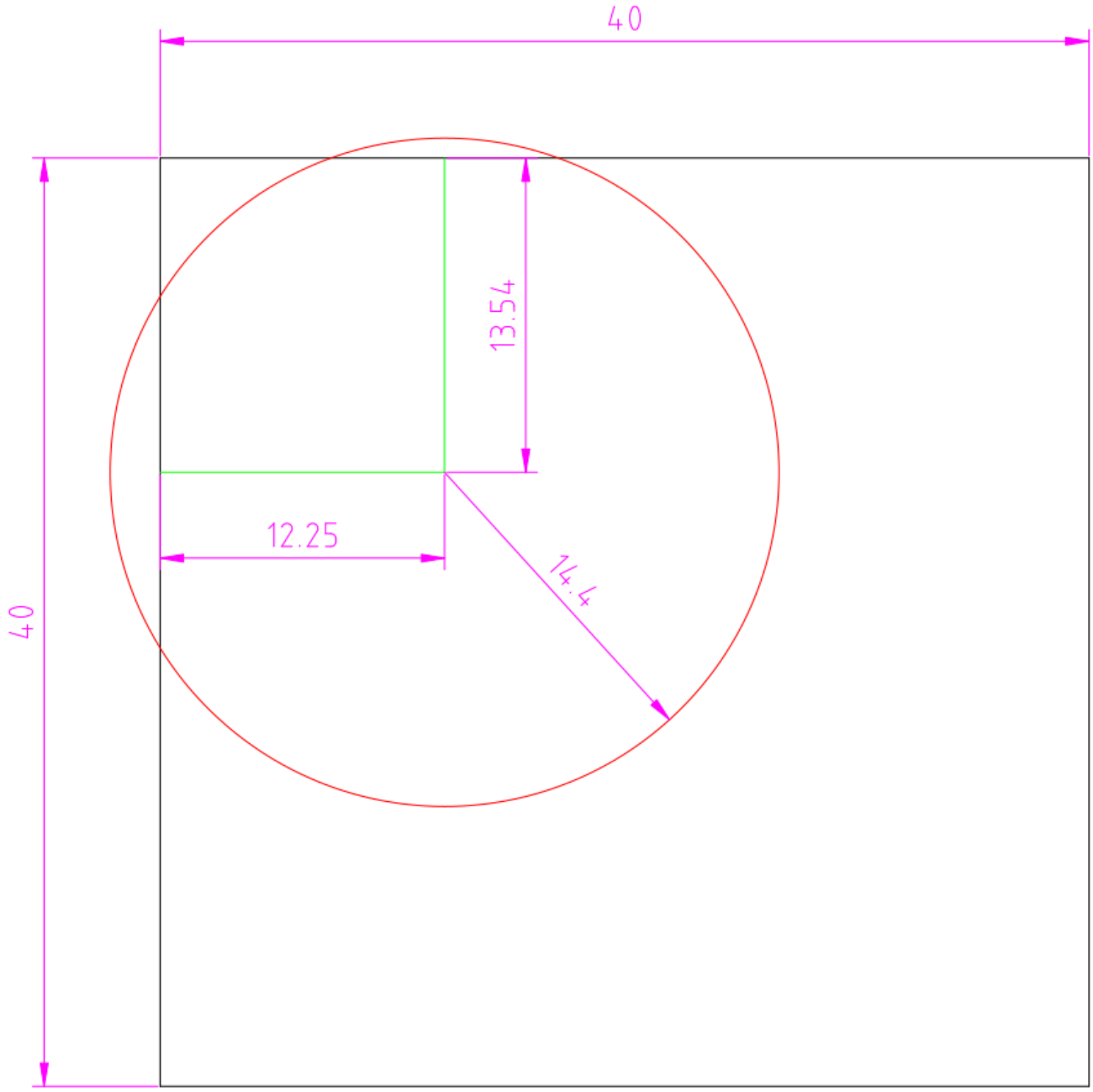


Figure 2: Sketch of Daedalus showing OOP field limitations

```
a.x.zero()
a.y.zero()
a.x.position = 12.25
a.y.position = 13.54
quit()
```

5. Wait for the Newport to stop moving before doing any experiments.

Motion Controller Stops Communicating

Another error that sometimes comes up is that the magnet will stop communicating with the computer all together. Fixing this is a bit more involved. Instructions are given below:

1. Restart the magnet by pressing the black power button twice. Wait for it to finish restarting. All of the axes will probably be disabled, noted by `off` being next to their readings on the controller.
2. open the Anaconda prompt
3. type `python` to open a Python interpereter
4. Type the following commands in the interpreter:

   ```
   from pymeasure.instruments.newport import ESP300
   a = ESP300(10)
   a.enable()
   ```

5. Check that all of the axes are properly enabled, noted by `on` being next to their measurements on the controller. If any are not on, try pressing the white buttons next to the axis which should turn them on. If that doesn't work, try running `a.enable()` again on the interpereter. If all else fails, restart the controller again.
6. We now need to re-zero the controller again. To do this, continuing from where we were on the interpereter before:

   ```
   a.x.position = -100
   a.y.position = -100
   ```

   after this, wait for the Newport to stop moving. It should beep to let you know that the X and Y stages have reached their negative hardware limits.

7. Now set the proper software zero:

   ```
   a.x.zero()
   a.y.zero()
   a.x.position = 12.25
   a.y.position = 13.54
   quit()
   ```

8. Wait for the Newport to stop moving before doing any experiments.

Motion Stages Stops Moving, Won't Respond

This happens sometimes, and the error which is returned is usually a tracking error. When this happens, it will not attempt to move and will probably be stuck at the positive hardware limit. Instructions to fix this are given below, assuming that the y axis is stuck:

1. Using your hand, *slightly* turn the screw controlling the y axis. We just want to get it back on track, so if it has moved too far positive twist the screw in the "-" direction, and vice versa for the other way. It should only move around 0.1 mm and may be quite hard to turn.
2. Follow the instructions for re-zeroing the magnet, although you may only need to run the steps for the affected axis. Use your judgment.